

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Tutorial Sheet 12:

11th and 12th May 2023

1. We introduced vector fields $\vec{F} : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (called two-dimensional vector) and written $\vec{F} = (P, Q)$ or $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ (called three-dimensional vector field) written as $\vec{F} = (P, Q, R)$, where P, Q, R are the component functions of the vector field. We saw few examples in the lectures, among which the notion of gradient vector field also called conservative vector fields, that is,

$$\begin{cases} \vec{F}(x, y) = \nabla f(x, y) & \text{for some function } f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}, \\ \vec{F}(x, y, z) = \nabla f(x, y, z) & \text{for some function } f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R} \end{cases}$$

That is, we seek $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\begin{cases} f_x(x, y) = P(x, y) \\ f_y(x, y) = Q(x, y) \end{cases} \quad (*)$

- Integrate the first equation w.r.t. x and get

$$f(x, y) = \int P(x, y) dx + C(y). \quad (**)$$

- Differentiate $(**)$ w.r.t. y and get $f_y(x, y) = \frac{d}{dy} \left[\int P(x, y) dx \right] + C'(y)$.

- Use the second equation in $(*)$ to get that $C'(y) = Q(x, y) - \frac{d}{dy} \left[\int P(x, y) dx \right]$.

- Find the function C and substitute it into $(**)$ to find the expression of the function f .

Similarly for 3D-vector fields!

Follow the steps above to show that each vector field below is conservative by finding a function f for which $\vec{F} = \nabla f$.

(i) $\vec{F}(x, y) = (3x^2 + 6y, 3y^2 + 6x);$

(ii) $\vec{F}(x, y, z) = (e^x \sin y \cos z, e^x \cos y \cos z, -e^x \sin y \sin z).$

2. We recall that if $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is a continuous function and C is a regular curve contained in D , described by a vector function $\vec{r} : [a, b] \rightarrow \mathbb{R}^2$, then the line integral of the function f along the curve C w.r.t. the arclength parameter s is given by

$$\boxed{\int_C f(x, y) ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| dt}$$

and similarly for $f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$.

Evaluate the line integrals:

- (a) $\int_C (x^2 + y^2) ds$ where C is the portion of the curve $x^2 + y^2 = 6y$ in the first quadrant joining $(0, 0)$ to $(3, 3)$.
- (b) $\int_C y ds$ where C is the arc defined by $y = \sqrt{x}$, $0 \leq x \leq 2$;
- (c) $\int_C z^2 ds$ and $\int_C xy ds$, where C is the curve described by the vector function $\vec{r} := \left[\frac{\pi}{2}, \frac{3\pi}{4}\right] \rightarrow \mathbb{R}^3$ defined by $\vec{r}(t) = (a \cos t, a \sin t, bt)$ for every $t \in \left[\frac{\pi}{2}, \frac{3\pi}{4}\right]$ with $(a, b) \neq (0, 0)$.

3. **(After the lecture on Friday)** Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}(x, y) := \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2}\right)$ and C is any circle centered at the origin and oriented clockwise.
Hint: First write down a parameterization of the curve C , that is, find a vector function describing the circle C with that orientation

4. **(After the lecture on Friday)** Let C_1 be the upper half of the ellipse $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$.
- (a) Sketch the curve C_1 .
 - (b) Suggest a parametrization of the curve C_1 .
 - (c) Let C be the union of the curve C_1 and the line segment L joining $(-2, 0)$ to $(4, 0)$. Evaluate the line integral $\int_C (x + 2y)dx + (2x + 3y)dy$ using the parametrization you suggested in (b).
 - (d) How else would you evaluate the integral in (c)?